High Performance Data Processing (SECP3133)

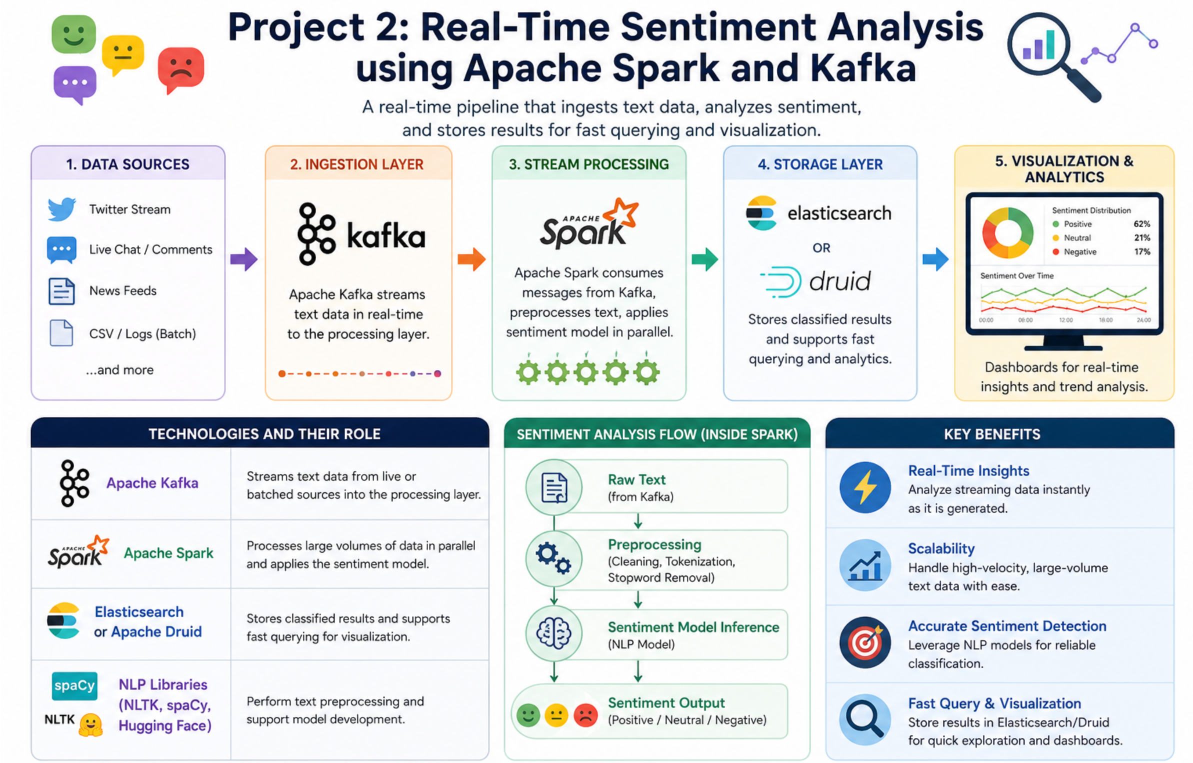
2025/2026, Semester 2

Project 2

Real Time Sentiment Analysis using Apache Spark and Kafka

Item	Details
Project Type	Group Project (Maximum 3 students per group)
Duration	4 Weeks
Weightage	15% of total course assessment
Submission Date	26 June 2026 (Friday)
Submission Channel	e-Learning Portal and GitHub Repository

1. Infographic



2. Introduction

Welcome to Project 2 of the High Performance Data Processing course. This project gives students a practical opportunity to design and build a real time sentiment analysis pipeline using modern big data technologies. The activities are designed to connect classroom theory with industry relevant data engineering practice.

In this project, each group will collect text data from Malaysian relevant sources, classify the sentiment of that text, and process it through a scalable pipeline built with Apache Kafka and Apache Spark. The final insights will be stored and presented through interactive dashboards.

This document is the official assignment brief. Students should read every section carefully before starting the project and refer back to it throughout the four week implementation period.

3. Project Overview

Organizations increasingly rely on real time analytics to understand public opinion about products, events, and social issues. This project simulates that environment by requiring each group to build a complete sentiment analysis system that operates on streaming data.

Each group will select a Malaysian relevant text source, such as social media posts, online reviews, or news articles. The collected text will be cleaned using natural language processing techniques, classified as positive, negative, or neutral by a trained model, and processed through a streaming pipeline. The resulting sentiment trends will then be stored and visualized so that insights can be communicated clearly to stakeholders.

3.1 Technologies Involved

The project is built around a set of widely adopted open source technologies. Each group is expected to integrate the following components into a working pipeline:

Technology	Role in the Pipeline
Apache Kafka	Streams text data from live or batched sources into the processing layer.
Apache Spark	Processes large volumes of data in parallel and applies the sentiment model.
Elasticsearch or Apache Druid	Stores classified results and supports fast querying for visualization.
NLP Libraries (NLTK, spaCy, Hugging Face)	Perform text preprocessing and support model development.

4. Learning Outcomes

Upon successful completion of this project, students will be able to:

a. Design and configure a real time streaming pipeline using Apache Kafka and Apache Spark.
b. Apply standard natural language processing techniques to clean and prepare text data.
c. Train, evaluate, and compare at least two sentiment classification models.
d. Integrate a trained model into a streaming system that classifies text as positive, negative, or neutral.
e. Store and visualize sentiment results using dashboard tools such as Kibana or Apache Superset.
f. Collaborate effectively in a small team and communicate insights through a professional report and presentation.

5. Project Requirements

5.1 Group Composition

This is a group based project. Students must form their own groups subject to the following requirements:

a. **Each group must consist of a maximum of three (3) students.** Groups with more than three members will not be accepted.
b. Each group must include members from different genders, races, or backgrounds in order to encourage diverse perspectives and inclusive collaboration.

c. Roles and responsibilities must be distributed clearly among members. Suggested roles include Group Leader, Data and NLP Engineer, and Pipeline and Visualization Engineer.

d. Group formation and the chosen data source must be confirmed with the lecturer no later than the end of Week 1.

5.2 Technical Requirements

The technical scope of the project is fixed. Each group must satisfy all of the following criteria:

a. Collect or stream text data from a Malaysian relevant source, such as Twitter, Shopee, Google Play reviews, or local news.

b. Preprocess the text using NLP methods, including cleaning, tokenization, stop word removal, and stemming or lemmatization.

c. Build and train at least two sentiment classification models, for example Naive Bayes and LSTM, and compare their performance.

d. Implement real time processing by streaming data through Apache Kafka and processing it with Apache Spark.

e. Store the classified output in Elasticsearch or Apache Druid and produce interactive visual dashboards.

f. Compare model and pipeline performance between batch mode and streaming mode.

5.3 Suggested Data Sources

Each group must select one primary data source. The data must be text based, publicly available, and suitable for classification into positive, negative, or neutral sentiment.

Category	Example Sources	Suggested Tools
Social Media	Twitter posts filtered by Malaysian keywords or hashtags	Tweepy, Twitter API v2
Customer Reviews	Shopee, Lazada, Google Maps, Google Play app reviews	BeautifulSoup, Selenium, Scrapy, Google Play Scraper
News Articles	The Star, Bernama, Free Malaysia Today RSS feeds	newspaper3k, RSS readers
Public Datasets	Sentiment140, IMDb, Amazon Reviews (for model training)	Kaggle, IEEE DataPort

Important note: Public labeled datasets may be used to train and validate models, but the final pipeline must process Malaysian relevant data. Confirm your chosen source with the lecturer before collection begins.

6. Technical Tasks

The project is organized into a four week schedule. Each week has clear objectives and milestones. Groups are encouraged to follow this schedule closely to ensure steady progress.

6.1 Weekly Schedule

Week	Phase	Key Activities
1	Planning and Data Acquisition	Form the group, select a data source, obtain lecturer approval, and begin collecting text data.
2	Preprocessing and Modeling	Clean and preprocess the text data, then build, train, and evaluate at least two sentiment models.
3	Pipeline Integration	Set up Apache Kafka and Apache Spark, integrate the chosen model, and connect the storage layer.
4	Visualization and Submission	Build dashboards, compare batch and streaming results, finalize the report, and submit all deliverables.

6.2 Suggested Role Assignments

Although roles may be adapted to suit each group, the following structure is recommended for a team of three:

Role	Primary Responsibilities
Group Leader and Data Engineer	Coordinates the team, manages data acquisition and preprocessing, and tracks progress against the timeline.
NLP and Model Engineer	Develops, trains, and evaluates the sentiment models, and prepares the model comparison.
Pipeline and Visualization Engineer	Builds the Kafka and Spark pipeline, configures storage, and develops the dashboards.

Note: Every member is expected to contribute to coding, testing, and documentation regardless of their primary role.

6.3 Model Development Requirements

Each group must develop and compare at least two sentiment classification approaches. The table below lists acceptable model categories:

Model Category	Examples and Tools
Machine Learning	Naive Bayes, Support Vector Machine, Logistic Regression (scikit-learn).
Deep Learning	LSTM, BiLSTM, GRU, or CNN (TensorFlow or PyTorch).
Transformer Models	BERT or DistilBERT (Hugging Face Transformers).

Models must be evaluated using accuracy, precision, recall, F1 score, and a confusion matrix. The dataset should be split into training, testing, and validation subsets, for example using a 70, 20, and 10 percent ratio.

6.4 Performance Comparison Requirements

To demonstrate the behavior of the system under different conditions, each group must compare the following between batch mode and streaming mode:

a. Total processing time required to classify a fixed volume of text.
b. Throughput, measured as the number of records processed per second.
c. Classification accuracy and consistency across both modes.
d. Resource usage, including CPU and memory, where measurable.

Results must be presented using clearly labeled tables and charts, accompanied by a short written interpretation.

7. Deliverables

Each group must submit the following five deliverables. All items will be assessed as part of the final grade.

No.	Deliverable	Description
1.	Final Report (PDF)	A complete technical report covering data acquisition, model development, system architecture, results, and findings. Must be uploaded to Turnitin in PDF format.
2.	Source Code	Python or Scala scripts, model files, and configuration files, organized clearly. Submit as a GitHub repository link or a ZIP archive.
3.	Dashboard and Dataset	A working visualization dashboard together with the exported cleaned dataset.
4.	Model Comparison	An evaluation of the different models, including batch versus streaming performance results.
5.	Presentation Slides	Slide deck for a clear and concise 10 minute group presentation.

7.1 Recommended GitHub Folder Structure

To maintain consistency and ease of grading, please follow the recommended folder structure when submitting your repository:

```
P2/[your_group]/
├── README.md
├── data/
│  ├── raw_data/
│  └── cleaned_data.csv
├── notebooks/
│  └── preprocessing.ipynb
├── model_training.ipynb
├── kafka_spark_pipeline/
│  ├── spark_streaming.py
│  ├── dashboard/
│  └── elastic_mappings.json
├── kibana_visualizations.json
├── reports/
│  └── final_report.pdf
├── presentation_slides.pptx
└── requirements.txt
```

Note: The structure above is presented for reference. In your actual repository, each indented line should appear as a separate folder or file.

8. Submission Guidelines

All submissions must follow the rules below. Late or incomplete submissions may result in a deduction of marks at the lecturer's discretion.

8.1 Submission Channels

Channel	Purpose
e-Learning Portal	Upload the final report (PDF), presentation slides (PPTX), and a link to your GitHub repository.
Turnitin	Upload the final report (PDF) for plagiarism verification.
GitHub Repository	Host all source code, notebooks, datasets, dashboard files, and configurations using the recommended folder structure.

8.2 Submission Deadline

All deliverables must be submitted by the following deadline:

Friday, 26 June 2026, before 11:59 pm (Malaysian Standard Time).

Each submission must clearly state the group name, the names of all members, and a contact email address.

8.3 Suggested Final Report Structure

The final report should follow a professional structure. The recommended sections are listed below:

a. Title Page, including project title, group name, member names, and matric numbers.
b. Table of Contents.
c. Introduction, covering background, objectives, and project scope.
d. Data Acquisition and Preprocessing, describing sources, tools, and cleaning steps.
e. Sentiment Model Development, including model choice, training process, and evaluation.
f. Apache System Architecture, with a workflow diagram for Kafka, Spark, and the storage layer.
g. Analysis and Results, presenting key findings, visualizations, and insights.
h. Optimization and Comparison, covering model and architecture improvements.
i. Conclusion and Future Work.
j. References.
k. Appendix, containing code snippets, configurations, and logs.

9. Evaluation Criteria

The project will be assessed using the rubric shown below. The total weightage is 100 percent of the project mark, which contributes 15 percent to the overall course grade.

Assessment Area	Description	Weight
Data Acquisition and Preprocessing	Quality of the data source, NLP cleaning steps, and dataset preparation.	15%
Sentiment Model Development	Correct training, evaluation, and comparison of at least two models.	25%
Apache Pipeline Integration	Working Kafka and Spark pipeline with correct model integration.	20%
Real Time Processing and Comparison	Effective streaming implementation and batch versus streaming comparison.	10%
Visualization and Dashboard	Clarity, accuracy, and usefulness of the sentiment dashboards.	10%
Final Report	Clarity, completeness, and overall professionalism of the document.	10%
Group Presentation	Teamwork, clarity of explanation, time management, and Q and A handling.	10%
Total	**Sum of all assessment areas.**	**100%**

10. Recommended Tools and Techniques

The following tools and libraries are widely used and well documented. Groups are not limited to this list, but it provides a strong starting point.

Category	Recommended Tools
Data Collection	Tweepy, BeautifulSoup, Selenium, Scrapy, Google Play Scraper, newspaper3k.
Text Preprocessing	NLTK, spaCy, Hugging Face Transformers, regular expressions.
Machine Learning Models	scikit-learn for Naive Bayes, SVM, and Logistic Regression.
Deep Learning Models	TensorFlow or PyTorch for LSTM, BiLSTM, GRU, and CNN.
Streaming and Processing	Apache Kafka, Apache Spark Structured Streaming (PySpark).
Storage	Elasticsearch, Apache Druid.
Visualization	Kibana, Apache Superset, Power BI, Tableau, Matplotlib, Seaborn, Plotly.
Deployment Support	Docker, Google Colab, Databricks, AWS EMR.

Suggested visualizations to include in the dashboard and report:

a. Sentiment distribution pie chart showing the proportion of positive, negative, and neutral entries.
b. Sentiment over time presented as a line graph.
c. Word clouds for frequent positive and negative terms.
d. A real time sentiment stream view, where the streaming setup allows it.

11. Example Workflow

The following workflow illustrates a typical end to end project execution. Groups may adapt these steps to suit their chosen data source and tools.

a. Select a data source. Identify a Malaysian relevant text source and confirm it with the lecturer.
b. Collect the data. Use an appropriate tool to gather a representative volume of text data.
c. Preprocess the text. Convert text to lowercase, remove noise such as URLs and mentions, and apply tokenization and lemmatization.
d. Build the models. Train at least two sentiment classifiers and evaluate them with standard metrics.
e. Select the best model. Choose the model with the strongest and most consistent performance for deployment.
f. Set up Kafka. Install Kafka, start the broker, and create a topic to receive streaming text.
g. Set up Spark. Configure a Spark streaming job that consumes data from the Kafka topic.

h. Integrate the model. Apply the trained model within the Spark job to classify incoming text in real time.
i. Store the results. Write the classified output to Elasticsearch or Apache Druid.
j. Visualize and analyze. Build dashboards, compare batch and streaming results, and interpret the findings.
k. Document and present. Compile the report, polish the slides, and rehearse as a team.

12. Common Mistakes to Avoid

Many groups encounter the same challenges. Being aware of them in advance can save valuable time.

a. Choosing a data source that is too small or not suitable for sentiment classification.
b. Skipping proper text preprocessing, which often reduces model accuracy significantly.
c. Training only one model, when the brief clearly requires at least two for comparison.
d. Treating the project as a model only task and neglecting the Kafka and Spark integration.
e. Leaving the streaming pipeline setup until the final week, when it usually needs the most debugging time.
f. Reporting accuracy numbers without explaining what they mean or why one model performs better.
g. Submitting code without comments, a clear folder structure, or an updated requirements file.

13. Tips for Success

a. Start data collection early, since gathering and labeling text data often takes longer than expected.
b. Test the Kafka and Spark setup with a small sample before processing the full dataset.
c. Use Docker or a cloud environment to reduce installation and configuration problems.
d. Keep the trained model files and configurations under version control.
e. Hold short weekly meetings to track progress and resolve blockers quickly.
f. Ask the lecturer for guidance whenever the scope or technical approach is unclear.
g. Always keep a backup copy of the dataset, models, and code outside the main repository.

14. Important Notes

Please pay close attention to the following points:

a. Plagiarism, including copying code or text without proper attribution, will be treated as a serious academic offense. Students who submit copied work may receive a mark of zero and face disciplinary action by the Faculty.
b. Each group member must contribute fairly. Imbalanced contributions may affect individual marks.

c. All data must be collected legally and ethically, and the terms of service of each source must be respected.

d. Submitting outside the official channels listed in this brief will result in the submission being treated as missing.

e. Any questions regarding scope or grading should be directed to the lecturer or teaching assistant during the course period.